

CITYCARE HOSPITALS · FORWARD DEPLOYED ENGINEERING

Discharge Copilot Design, made simple

How the old hospital system works, the POC we built, and how it all connects

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Colours in diagrams: **Old system** **New Copilot** **AI** **People**

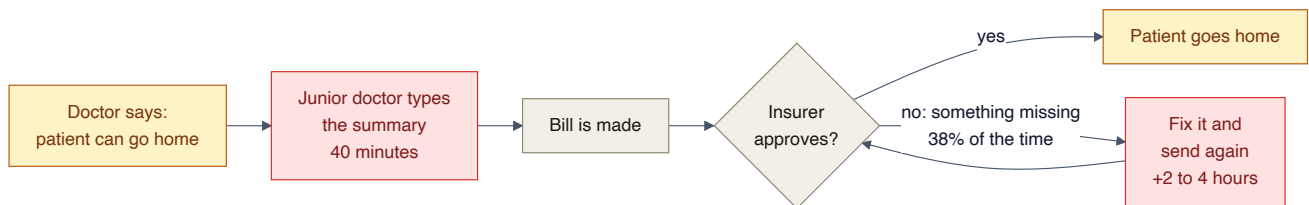
CityCare Hospitals, its patients, doctors and numbers are fictional and used for teaching.

1. The problem

Why insured patients wait so long to go home.

IN SIMPLE WORDS

When a doctor says a patient can go home, someone has to write a **discharge summary** (a report of the whole stay). For insured patients, the insurance company must also approve the bill, and it needs that summary. Writing it is slow, and it often misses things the insurer wants, so the insurer sends it back.



What happens on the day a patient leaves. The red boxes are where time is lost.

6.3 hours

wait for insured patients
(cash patients: 2.5 hours)

38%

of insurance submissions
are sent back

40 min

to type one summary

Our solution in one line

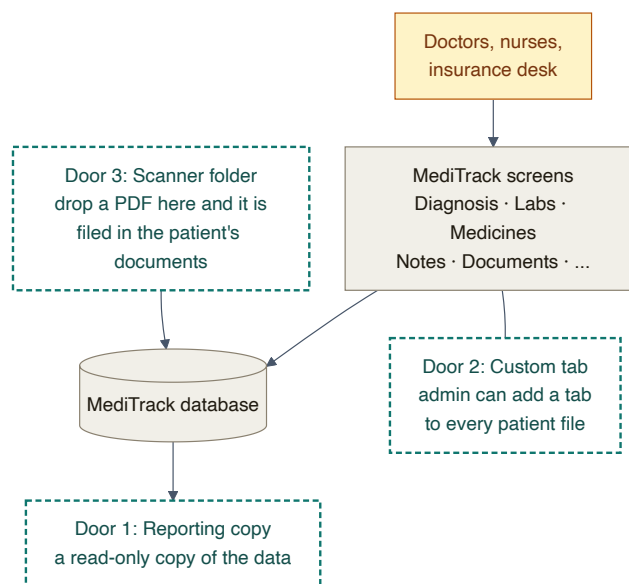
The moment the doctor says "go home", **AI writes a first draft** of the summary, **a checklist flags anything the insurer will ask for**, and **the doctor reviews and signs it**, all inside the hospital's existing system.

2. The old system (MediTrack)

The hospital software we must work with, not replace.

IN SIMPLE WORDS

MediTrack was built in 2011. The company that made it no longer exists, so **nobody can change its code**. Patient information is spread across many screens. But it has three "doors" we can use safely without touching its code.



MediTrack and its three safe doors (dashed boxes).

What MediTrack stores (the parts we use)

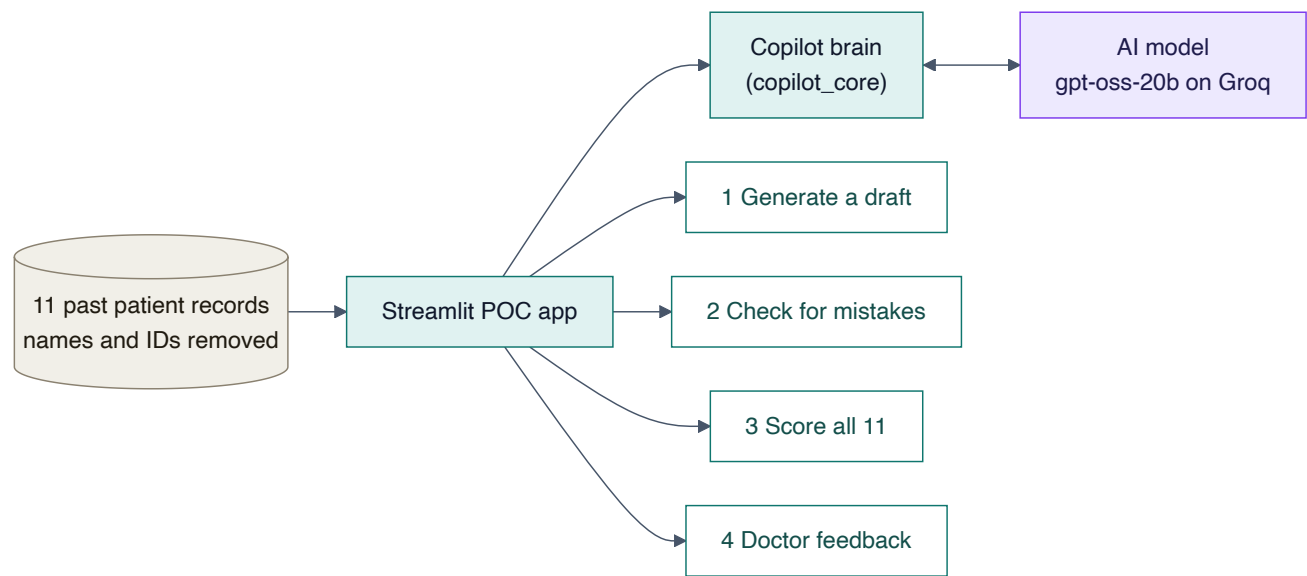
Information	Example for our demo patient, Ramesh Kulkarni
Patient and admission	62 years, admitted 21 Sep with a heart attack, insured (cashless)
Diagnosis	Heart attack (ICD-10 code I21.0), high blood pressure
Procedures	Angiography and stent placed in a heart artery
Lab results	Troponin 12.4, heart pumping at 40%, one report still pending
Medicines	7 medicines to take home
Doctor and nurse notes	Free-text notes written each day
Documents	Consent, cath report, but the stent sticker is missing (insurers always ask for it)

3. The POC: high-level design

A small test app to prove the AI is safe and useful before touching the real system.

IN SIMPLE WORDS

We took **11 old patient records**, removed names and IDs, and built a simple app in **Streamlit**. It lets us generate AI drafts, check them for mistakes, score them against what the doctors actually wrote, and collect doctors' opinions. It is **not connected** to the hospital system.



The POC and its four screens.

The most important design rule

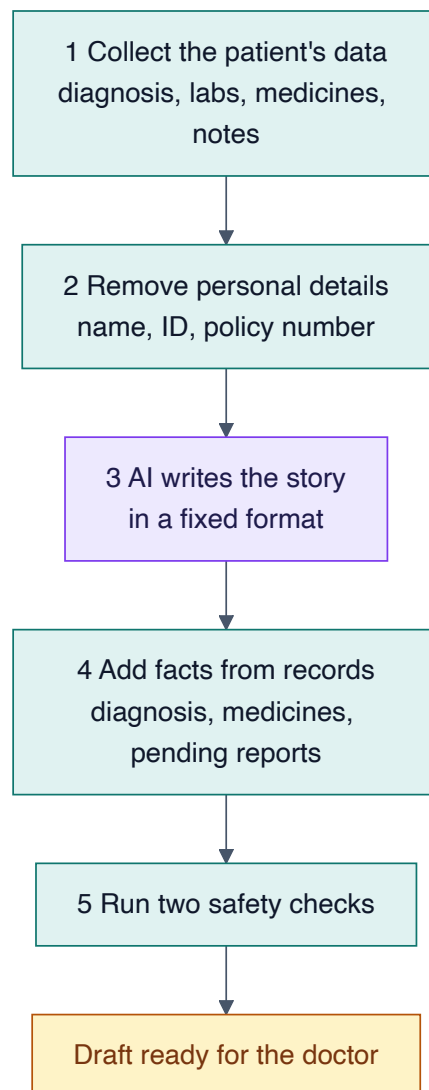
Part of the summary	Who writes it
Diagnosis, medicines, pending reports	Copied straight from hospital records. Never written by AI, because a mistake here could harm a patient.
Insurance checklist	Simple rules in code. We already know what insurers ask for.
The story of the stay: symptoms, what happened each day, test highlights, condition at discharge, follow-up	AI writes the draft. Turning messy notes into a clear story is what AI is good at. The doctor can edit everything.

Tools used

Streamlit (screens) · **Pydantic AI** (talks to the AI and makes it return a fixed format) · **Groq gpt-oss-20b** (the AI model) · **Logfire** (records every AI call so we can inspect it)

4. The POC: how it works inside

What happens when you click "Generate".



Five steps from patient data to a draft summary.

The two safety checks

Check	What it does
Fact check	Every medicine, test name and number in the AI text is looked up in the patient's record. Green = found. Yellow = number not found. Red = not in the record at all (the AI made it up).
Insurance checklist	7 simple rules, e.g. "is there a final diagnosis?", "is the stent sticker attached?", "are pending reports mentioned?". Gives a readiness score such as 71%.

If the AI is not available

The app never shows an empty screen. It tries, in order: **live AI** → a **saved copy** of an earlier AI draft → a **basic draft** built directly from the notes.

Results on the 11 test patients



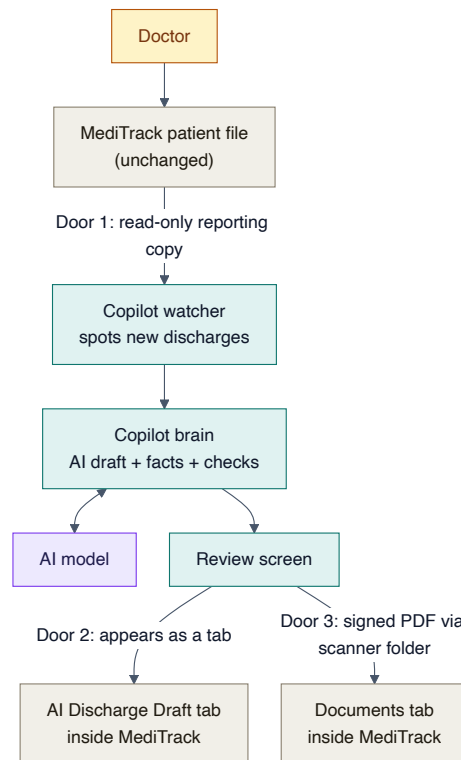
What we learned: the AI once wrote wrong dates, and the fact check caught them. Follow-up advice is often missing because doctors don't write it in their notes. That's a process fix, not an AI fix.

5. The integrated system: high-level design

The real thing, running next to MediTrack.

IN SIMPLE WORDS

The Copilot runs **beside** MediTrack and connects only through the three safe doors. It **reads** from the reporting copy, **shows** the draft in a new tab inside MediTrack, and **sends back** the signed summary through the scanner folder. MediTrack's code is never changed, and doctors never leave MediTrack.



How the Copilot connects to MediTrack through the three doors.

Why a tab, not a new app

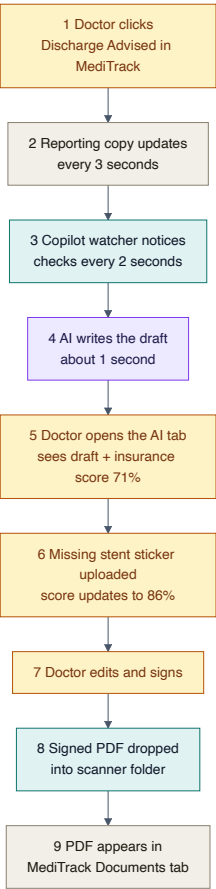
300+ doctors and nurses already know MediTrack. A new app means new logins, training and resistance. A tab that **looks exactly like MediTrack** (same grey style, same buttons) means the only new habit is: *"after Discharge Advised, open the AI tab."*

Built-in safety

- **MediTrack is never written to directly:** the Copilot only reads the copy and uses the scanner folder.
- **A doctor must sign:** nothing reaches the patient or insurer without it.
- **Personal details are removed** before anything is sent to the AI.
- **Every step is logged**, including who edited and who signed.

6. The integrated system: how it works inside

One patient, start to finish, in about 10 seconds of system time.

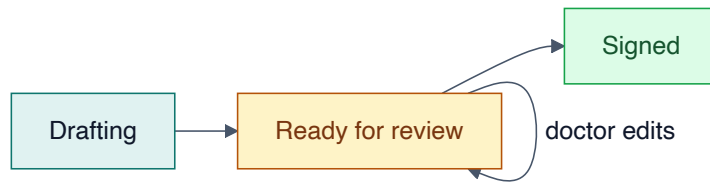


The full journey for Ramesh Kulkarni.

The main parts

Part	Job
Reporting copy	A read-only copy of MediTrack's data, refreshed every 3 seconds
Watcher	Looks for patients marked "Discharge Advised" and starts a draft
Copilot brain	The same code as the POC: AI draft + facts from records + two safety checks
Copilot database	Keeps each draft, the doctor's edits, and a log of every action
Review screen	Shown inside MediTrack's tab: edit, check, sign
PDF writer	Creates the signed summary and drops it in the scanner folder
Monitor page	For management: before vs after numbers and activity

A draft's life



If something goes wrong

Problem	What happens
AI or internet is down	A saved draft or a basic draft is shown instead
Copilot is down	MediTrack keeps working; doctors use the old Word template
A document is still missing	The checklist warns the doctor before signing

7. Summary

Everything on one page.

	Old system	POC	Integrated system
What it is	MediTrack, 2011 hospital software	Test app on 11 anonymised past patients	Copilot running beside MediTrack
Purpose	Stores all patient data	Prove the AI is safe and useful	Save time for real patients
Built with	FastAPI, SQLite, React	Streamlit, Pydantic AI, Groq, Logfire	FastAPI, React, same AI brain as the POC
Connected to hospital?	It is the hospital system	No	Yes, through 3 safe doors only
Who uses it	All staff	Project team, doctors (for feedback)	Doctors (AI tab), managers (monitor)

THE BIG IDEA

AI writes the story. The records supply the facts. Simple rules check what the insurer needs. **The doctor stays in charge and signs.** And the old system keeps running exactly as before.